

Week 4 - Vertical Profiles: Worked Answers

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

The categories are repeated here so the worked answers map directly onto the practice sheet. Use the solutions to check method, units and physical interpretation rather than only the final number. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Unless stated otherwise, use $g = 9.81 \text{ m s}^{-2}$, $R_d = 287 \text{ J kg}^{-1} \text{ K}^{-1}$ and a representative pressure scale height $H = 8.2 \text{ km}$.

IN CLASS Question 1: Deriving hydrostatic balance

Consider a horizontal slab of air of area A , thickness dz and density ρ .

1. Draw and label the vertical pressure forces and the weight of the slab.
2. Derive the hydrostatic equation

$$\frac{dp}{dz} = -\rho g.$$

3. State clearly what physical balance the equation represents.

Worked answer

Pressure on the lower face produces an upward force $p(z)A$. Pressure on the upper face produces a downward force $p(z + dz)A$. The weight is $\rho A dz g$ downward.

For no vertical acceleration,

$$p(z)A - p(z + dz)A - \rho A dz g = 0.$$

Rearrange and divide by $A dz$:

$$-\frac{p(z + dz) - p(z)}{dz} = \rho g.$$

Taking the limit $dz \rightarrow 0$ gives

$$\frac{dp}{dz} = -\rho g.$$

The upward pressure-gradient force balances the downward gravitational force. The atmosphere can still contain small vertical motions; “hydrostatic” means that vertical acceleration is small enough that this balance is a good approximation.

EXTRA CHALLENGE Question 2: Pressure levels and atmospheric mass

At sea level take $p_s = 1013$ hPa. Under hydrostatic balance, pressure at a level is proportional to the mass of atmosphere above that level per unit area.

1. What fraction of the total atmospheric mass lies above the 850 hPa surface?
2. Repeat for 350 hPa and 10 hPa.
3. Explain why these results are useful for interpreting pressure coordinates in meteorology.

Worked answer

The mass above a pressure surface per unit area is p/g , so relative to the mass above sea level the fraction is simply p/p_s .

$$\frac{850}{1013} = 0.839, \quad \frac{350}{1013} = 0.346, \quad \frac{10}{1013} = 9.87 \times 10^{-3}.$$

Thus about 84%, 35% and 1% of the atmosphere respectively lie above those pressure levels. Pressure coordinates therefore have a direct mass interpretation: moving upward to lower pressure means that less atmospheric mass remains above you.

YOU SHOULD TRY Question 3: How high is the 500 hPa level?

Assume

$$p = p_0 \exp(-z/H),$$

with $p_0 = 1013$ hPa and $H = 8.2$ km. Estimate the altitude of the 500 hPa surface.

Worked answer

Rearrange the exponential relation:

$$z = -H \ln \left(\frac{p}{p_0} \right).$$

Hence

$$z = -8.2 \ln \left(\frac{500}{1013} \right) = 5.79 \text{ km}.$$

The 500 hPa surface is therefore at roughly 5.8 km in this simple atmosphere. Its actual height varies with the temperature of the atmospheric column.

EXTRA CHALLENGE Question 4: Density with height

Assume $\rho_0 = 1.225 \text{ kg m}^{-3}$ and the same scale height $H = 8.2$ km. Calculate the density at 5 km and 10 km using

$$\rho = \rho_0 \exp(-z/H).$$

Worked answer

At 5 km,

$$\rho = 1.225 \exp(-5/8.2) = 0.666 \text{ kg m}^{-3}.$$

At 10 km,

$$\rho = 1.225 \exp(-10/8.2) = 0.362 \text{ kg m}^{-3}.$$

The large reduction in density with height affects aircraft lift, human physiology and the interpretation of atmospheric measurements.

YOU SHOULD TRY Question 5: Aircraft altimetry and pressure errors

An aircraft is actually at 9.0 km in an atmosphere described by $p = p_0 e^{-z/H}$ with $p_0 = 1013$ hPa and $H = 8.2$ km.

1. Estimate the ambient pressure.
2. Two aircraft are intended to remain vertically separated by 300 m. Approximately what pressure difference at 9 km corresponds to a 300 m altitude difference in this model?
3. Why does this illustrate the importance of pressure calibration in barometric altimetry?

Worked answer

At 9.0 km,

$$p = 1013 \exp(-9/8.2) = 338 \text{ hPa.}$$

At 9.3 km the pressure is

$$p_{9.3} = 338 \exp(-0.3/8.2) = 325.9 \text{ hPa.}$$

The difference is therefore about

$$338.0 - 325.9 = 12.1 \text{ hPa.}$$

A pressure error of only order 10 hPa at this altitude can correspond to hundreds of metres in inferred altitude. Real aviation altimetry uses standardised procedures and corrections precisely because pressure-to-height conversion is safety critical.

YOU SHOULD TRY Question 6: Reducing a hill-station pressure to sea level

A barometer reads 950 hPa at a station 500 m above sea level. Assume an isothermal layer at 273 K.

1. Calculate the scale height $H = R_d T / g$.
2. Estimate the equivalent sea-level pressure using $p(z) = p_0 e^{-z/H}$.
3. Name two reasons why a real meteorological sea-level pressure reduction requires more care than this calculation.

Worked answer

The scale height is

$$H = \frac{R_d T}{g} = \frac{287 \times 273}{9.81} = 7.99 \times 10^3 \text{ m} = 7.99 \text{ km.}$$

Since the station is at $z = 0.5$ km,

$$p_0 = p(z) \exp(z/H) = 950 \exp(0.5/7.99) = 1011 \text{ hPa.}$$

Real reductions need a representative temperature for the air column between station and sea level, not simply the station temperature. Instrument calibration, local gravity and exposure/dynamic pressure effects can also matter.

IN CLASS Question 7: Lapse rate and dry stability

A sounding gives the following temperatures in the lowest 3 km:

Height (km)	0	1.0	2.0	3.0
Temperature (°C)	18	12	7	2

1. Calculate the mean environmental lapse rate over 0–3 km.
2. Compare it with the dry adiabatic lapse rate, $\Gamma_d \approx 9.8 \text{ K km}^{-1}$.
3. Would an unsaturated parcel displaced upward through this layer tend to become warmer or colder than its environment? What does that imply for dry static stability?

Worked answer

The environmental temperature falls by 16 K over 3 km, so

$$\Gamma_{env} = \frac{16}{3} = 5.33 \text{ K km}^{-1}.$$

This is substantially smaller than the dry adiabatic lapse rate of 9.8 K km^{-1} . An unsaturated rising parcel cools faster than the environment, so after an upward displacement it becomes colder and denser than its surroundings. It tends to sink back: the layer is stable to dry convection.

EXTRA CHALLENGE Question 8: Scale height on Earth and Mars

Use

$$H = \frac{R^*T}{Mg}$$

with $R^* = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.

1. For Earth use $T = 288 \text{ K}$, $M = 28.97 \times 10^{-3} \text{ kg mol}^{-1}$, $g = 9.81 \text{ m s}^{-2}$.
2. For Mars use $T = 210 \text{ K}$, $M = 44.01 \times 10^{-3} \text{ kg mol}^{-1}$, $g = 3.71 \text{ m s}^{-2}$.

Calculate both scale heights and explain why Mars can have a larger scale height despite its colder atmosphere.

Worked answer

For Earth,

$$H_E = \frac{8.314 \times 288}{(28.97 \times 10^{-3}) \times 9.81} = 8.43 \text{ km}.$$

For Mars,

$$H_M = \frac{8.314 \times 210}{(44.01 \times 10^{-3}) \times 3.71} = 10.7 \text{ km}.$$

Mars is colder and its CO_2 atmosphere has a larger molar mass, both of which reduce scale height. However, Martian gravity is much weaker than Earth's, and that effect is large enough to produce the greater scale height in this comparison.

EXTRA CHALLENGE Question 9: Hydrostatic pressure in an atmosphere with a constant lapse rate

Assume hydrostatic balance, ideal-gas air, and a temperature profile

$$T(z) = T_0 - \Gamma z,$$

where $T_0 = 288$ K and $\Gamma = 6.5$ K km⁻¹. Starting from

$$\frac{dp}{dz} = -\rho g, \quad p = \rho R_d T,$$

show that

$$p(z) = p_0 \left(\frac{T(z)}{T_0} \right)^{g/(R_d \Gamma)}.$$

Use $p_0 = 1013$ hPa to estimate the pressure at $z = 5.0$ km and compare it with the isothermal scale-height estimate using $H = 8.2$ km.

Worked answer

Combining hydrostatic balance with the ideal-gas law gives

$$\frac{dp}{p} = -\frac{g}{R_d(T_0 - \Gamma z)} dz.$$

Integrating from $(p_0, 0)$ to (p, z) ,

$$\ln \frac{p}{p_0} = \frac{g}{R_d \Gamma} \ln \frac{T_0 - \Gamma z}{T_0},$$

so

$$p(z) = p_0 \left(\frac{T(z)}{T_0} \right)^{g/(R_d \Gamma)}.$$

At 5 km, $T = 288 - 6.5 \times 5 = 255.5$ K and, using $\Gamma = 0.0065$ K m⁻¹,

$$\frac{g}{R_d \Gamma} = \frac{9.81}{287(0.0065)} = 5.26.$$

Hence

$$p = 1013 \left(\frac{255.5}{288} \right)^{5.26} \approx 540 \text{ hPa}.$$

The isothermal estimate is

$$p = 1013 e^{-5/8.2} \approx 551 \text{ hPa}.$$

The two are similar at 5 km, but the lapse-rate solution is more physically representative because temperature decreases with height.

Modelling connection.

The following problems connect vertical-profile physics to two useful modelling ideas: conservation of dry potential temperature during adiabatic motion, and the onset of rain by collision-coalescence. Students who also take EART22001 will meet related ideas in a single-column model, but no knowledge of that unit is assumed here. The detailed cloud microphysics is revisited in Week 9.

IN CLASS Question 10: Why potential temperature is a useful conserved variable

A dry parcel begins at $p_1 = 950$ hPa and $T_1 = 285$ K. It is displaced adiabatically to $p_2 = 800$ hPa.

1. Calculate its initial potential temperature using $\theta = T(1000/p)^{0.286}$.
2. Assuming θ is conserved, calculate the parcel temperature at 800 hPa.
3. Explain why a numerical atmospheric model may predict or conserve θ rather than temperature directly for dry adiabatic motion.

Worked answer

Initially,

$$\theta = 285 \left(\frac{1000}{950} \right)^{0.286} = 289.2 \text{ K.}$$

Conservation of θ gives

$$T_2 = \theta \left(\frac{800}{1000} \right)^{0.286} = 289.2(0.8)^{0.286} = 271.3 \text{ K.}$$

Temperature changes simply because pressure changes during adiabatic expansion/compression, whereas dry potential temperature removes that pressure dependence. It is therefore a natural thermodynamic state variable for diagnosing and modelling adiabatic atmospheric motion.

YOU SHOULD TRY Question 11: A first look at collision-coalescence and rain formation

Two cloud droplets have radii $10 \mu\text{m}$ and $25 \mu\text{m}$. In the Stokes regime take terminal speed to scale as $v_t \propto r^2$.

1. By what factor does the larger droplet fall faster than the smaller one?
2. Explain how this differential fall speed can produce collisions.
3. If every collision led to coalescence, would all droplets remain the same size? Explain why a broadened droplet spectrum helps rain formation accelerate.

Worked answer

The speed ratio is

$$\frac{v_{25}}{v_{10}} = \left(\frac{25}{10} \right)^2 = 6.25.$$

The larger drop therefore overtakes smaller drops below it. Some encounters lead to collision and, if the droplets merge, the collector becomes still larger. Once a size difference has developed, larger drops have both greater fall speeds and larger sweep-out areas, so collection can accelerate. This is why collision-coalescence tends to broaden the size distribution rather than keep it narrow.